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Title of the article: *Tracking tumor biology with radiomics: A systematic review utilizing a radiomics quality score*

The content extracted is as follows:

Radiomic studies may help to characterize tumor biology in vivo by correlating imaging features with ground truth pathology substrates.

Although medical images are commonly used by radiologists to describe a pathologic substrate, features extracted in the radiomic process are not part of radiologists' lexica since they are generally not visible to the naked eye.

In this review recent developments in the field of radiomics are described, along with relevant literature linking it to tumor biology.

General comments and recommendations for the improvement of this tool are presented. Finally, we offer our vision on necessary future steps for the development of stable and robust radiomic biomarkers and their link to tumor biology.

Materials and methods

The research question for this systematic review is described as: What are the known studies linking radiomics to tumor biology and what is the methodological quality of these studies?

We used a search string containing free-text and/or Medical Subject Headings (MeSH) search of 3 key search terms: biology, neoplasms and radiomics while search terms such as imaging features were not used as our purpose was to review manuscripts using more recent, high throughput technology.

From the included articles, data concerning study characteristics was extracted and tabulated (author, publication year, study population, radiomic feature extraction, main results and conclusion) together with measurement characteristics (i.e., method for quantifying biological mechanism).

Results

By basing these clinical associations exclusively on radiomic features, they could demonstrate that the associated molecular pathways robustly matched radiomic-based hypotheses.

A total of 17 studies established a correlation between cancer imaging features and a certain gene expression or gene mutation, a process called radiogenomics [12–28].

In this preclinical study gene induction with doxycycline (dox+) and irradiation translated into significant changes in radiomic features in GADD34 mouse xenografts (between the dox+ and irradiation groups 8 and 4 of these features remained consistent for 40 and 80kVp, respectively).

A total of nine studies associated radiomic features with tumor histopathological grading/malignancy prediction [29–37].

Another 15 studies explored the correlation between imaging features and a specific histological substrate, e.g., hormone receptor status or microvascular density (MVD) [37–51].

Finally, three studies found correlations between tumor metabolism and radiomic features [22,51,52].

Discussion

In this study, we performed a systematic literature review of studies addressing the interplay between radiomic features and tumor biology.

In total 39 out of 41 studies reveal that radiomic features derived from mammography, ultrasound, CT, PET and/or MRI are significantly associated with one or several specific tumor biologic substrates, from somatic mutation status to tumor histopathological grading and tumor metabolism.

The RQS score mentions the use of a biological correlate in a specific study, but does not quantify nor puts in perspective the link between tumor biologic substrate and radiomics.

A good example here of the contrast between work quality and impact is a study performed by Grossmann et al. [13] which was assigned less than half of the RQS points (mean = 38.9%) e.g., due to not adequately reporting the image protocol and not addressing the cut-off values used for the different biologic endpoints while on the other hand exhibiting a sound methodology (e.g., feature reduction, external validation) and performing a comprehensive genetic profiling of the tumors on a relatively large study population (n = 351).

The two major types of features extracted in the radiomic process are semantic and agnostic [54].

Some recent studies have focused on semantic features, which are qualitative radiologic features, e.

g., presence of cavitation, location of the lesion, presence of pleural effusion and shape of the lesion [22,52,55-57].

These studies found significant differences in the semantic features between mutated tumors and non-mutated tumors.

Both types of radiomic features convey pertinent information that is complementary to each other.

In contrast, agnostic radiomic features can be swiftly extracted with minimal operator input and make up the bulk of highly mineable quantitative data.

A key hypothesis driving radiomics research is the potential to enable a spatiotemporal and quantitative measurement of both intra- and intertumoral heterogeneity based on medical images.

The majority of the studies presented fall within the field of radiogenomics; genomic heterogeneity within tumors and across metastatic tumor sites in the same patient is currently thought to be a major cause of treatment failure and the emergence of (targeted) therapy resistance [58].

Aside from (epi)genetic heterogeneity, a complex and highly heterogeneous synergistic interplay exists between tumor cellular (proteomic, metabolic, morphological) phenotype and the microenvironment surrounding the tumor bed. Several studies investigated quantitative CT/PET-derived features associated with genomic and proteomic tumor expression [12,28,35,37,47,49,51].

One study (Yip et al.) found that EGFR mutated tumors shows different metabolic tumor phenotypes on PET than KRAS mutated tumor [22].

Another study (Aerts et al.) found that baseline radiomic features were able to predict mutation status and were also able to predict tyrosine kinase inhibitor (TKI) treatment response [27].

The supplementary value of radiomic to current genomic biomarkers is well illustrated in a study (Emaminejad et al.) where a quantitative image feature-based classifier yielded significantly higher discriminatory power than a genomic biomarker-based (AUC of 0.78 ± 0.06 and 0.68 ± 0.07 respectively) classifier in predicting early stage lung cancer recurrence risk [25].

In window-of-opportunity trials, radiomic features extracted from images before (and after) receiving an experimental drug and prior to the start of conventional treatment can potentially serve as predictive biomarker for clinical endpoints such as progression-free survival and (early) tumor response [61,62].

A recent study shows the predictive value of radiomic features extracted from weekly CT images for NSCLC patient outcomes [60].

The most predictive radiomic marker (a gray-level cooccurrence matrix (GLCM) feature) on the other hand differentiated low grade (LG) and high grade (HG) lesions better than the Fukuda criteria (21% incorrectly labeled for surgery) with an AUC of 0.82 at a sensitivity of 85% and specificity of 68%.

Even when adequate cytology and/or surgical biopsy specimens are available, radiomics offers important advantages for the assessment of tumor biology.

Radiomics could be a potential candidate to capture the temporal changes of such biological processes over various tumor habitats.

Several components are not always applicable to every study, e.g., clinical utility for pre-clinical studies or comparison to gold standard for outcomes in which no gold standard exists for the assessment of the outcome (e.g., tumor heterogeneity).

Our vision for the future of radiomics is intrepid, acknowledging that fundamental questions still need to be addressed before clinical applicability of radiomic imaging biomarkers will be achieved. With the advent of pre-clinical inducible knockout system studies, whereby a specific target gene of interest can be inactivated at a given time point, researchers will hopefully be able to draw more robust conclusions in the so-far mainly correlative relationship of radiomic features with tumor biology [26].

To find imaging biomarkers with stronger causal rather than correlative link with biological mechanisms would in our vision require not only radiomic studies on a vast amount of ultra-high resolution images to present high discrimination statistics in multiple validation sets, but also benchmarking and standardization of image preprocessing and feature extraction and data-sharing techniques that exploit matching ontologies.